TitanTPP: Local and Recurrent History Correction for Event Quantity Prediction

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Abstract. Event forecasting requires predicting how much will occur as well as when it will occur. A history encoder supplies information for both predictions, but this does not establish whether a separate path through recent observed states is useful. We study this question in TitanTPP, a causal attention model with an explicit correction path. The representative architecture applies a three-position correction to queries, keys, and values, then adds a gated recurrent residual between attention blocks. A separate comparison examines prototype retrieval with time and quantity queries. In a completed 54-run study of an earlier MLP correction, adding the correction reduced validation quantity RMSE in all 18 paired comparisons across three datasets. The benefit was largest on Taxi, where mean RMSE fell by 13.0–14.1%, but recorded-time likelihood deteriorated. Prototype addition was less consistent. These findings motivate controls that separate previous-state input from additional parameters, and split-query retrieval from general query learning. The CNN+GRU controls remain under evaluation. We report the completed MLP comparison and one supplementary Instacart fit as development-validation evidence.

Keywords: Event forecasting · Quantity prediction · History correction · Temporal point processes

1 Introduction

Intermittent demand alternates between periods without demand and nonzero observations whose sizes can vary. Forecasting therefore requires an account of both the interval to the next demand and its quantity. Croston's method separates demand size from the interval between nonzero observations [5]; later renewal models place this event-based description within a probabilistic framework [24].

Temporal point processes provide a natural representation of event arrivals. Several neural formulations, including RMTPP and Transformer Hawkes Process, predict categorical event types alongside their occurrence times [7,28]. Demand quantity has a numerical magnitude, however, and a type-classification objective does not directly account for the size of a quantity error. The distinction concerns categorical types versus numerical values, rather than discrete versus continuous data: demand counts can themselves be integer-valued. Point processes are not restricted to categorical marks. Deep renewal models already address demand size and arrival intervals, while mixed-attribute event models support numerical attributes [24,6].

Our question concerns the representation of observed history within this numerical prediction task. Recurrence, attention, and continuous-time states already summarize

earlier events [15,26,3]. Yet this does not determine whether a separate path through recent observed states improves quantity prediction. We ask whether an explicit correction adds useful structure beyond a history-aware encoder and a control with the same active parameter budget. All arms receive numerical event inputs and share quantity and timing output heads, so their comparison concerns the correction and retrieval paths.

We investigate this question with TitanTPP. The representative CNN+GRU54 model combines two forms of correction. A local path adjusts attention queries, keys, and values using the current and two preceding observed positions. A recurrent path adds a projected GRU state between the two attention blocks. The suffix 54 denotes the GRU state width. Both paths are residual, allowing the initial model to retain the shared encoder computation. They provide a particular way to reuse history, rather than supplying information unavailable to the causal encoder.

An earlier MLP version gives a reason to examine the correction more closely. Across Taxi, Intermittent, and RAF Spare Parts, switching on its explicit previous-state path improved overall quantity RMSE in all 18 comparisons formed by three seeds and two prototype settings. Yet the strongest quantity gains came with worse timing likelihood on Taxi. Moreover, adding prototypes to the corrected model did not improve quantity RMSE consistently across datasets. This pattern motivates two separate questions: the value of an explicit history path, and the value of the rule used to retrieve from a learned prototype bank.

The distinction matters for attribution. Removing the history path also removes parameters, so that comparison alone cannot identify the benefit of its input. We therefore include a control with the same number of active correction parameters that operates only on the current position. For retrieval, we compare uniform averaging, similarity weighting, separately normalized time and quantity queries, and a general learned query. Only the last two introduce query parameters, and their budgets are matched. Each study uses the same encoder architecture, output-head definitions, and model-selection rule; each arm is trained independently. Quantity is the primary selection objective, while timing likelihood, large-quantity errors, and computational cost describe the accompanying tradeoffs.

This paper makes three contributions:

1. We describe a local and recurrent correction architecture for next-event quantity prediction within a causal attention model, including the observation rules that prevent access to the target event.
2. We formulate controlled comparisons for explicit history input and prototype retrieval, separating these questions from extra parameter count and from the choice of output distribution.
3. We report the completed MLP correction study, including seed consistency, large-quantity errors, and timing tradeoffs. Its results support further testing of explicit correction, while showing why quantity improvements alone do not establish a uniformly better event model.

The empirical analysis in this version concerns the MLP study. The representative CNN+GRU comparisons and the supplementary Instacart architecture comparison are specified separately, so that a result for one architecture is not treated as evidence for another.

2 Related Work

Intermittent demand and numerical quantities. Croston models nonzero demand sizes and interdemand intervals separately [5]. The Syntetos–Boylan adjustment addresses bias in its demand-rate estimate [22], whereas Teunter, Syntetos, and Babai model occurrence probability to respond to the risk of inventory obsolescence [23]. Deep renewal processes extend the size–interval representation with learned conditional distributions [24]. DeepAR instead learns autoregressive probabilistic forecasts across related time series [17]. These approaches establish numerical demand prediction as an existing research problem. Our target is the next event's quantity and recorded gap; forecasting demand accumulated over a fixed calendar horizon would require an additional aggregation procedure.

Neural event-history encoders. Hawkes processes describe self-excitation and mutual excitation in event arrivals [9]. Neural TPPs replace fixed history summaries with learned representations and conditional distributions [19]. RMTPP uses a recurrent embedding [7], and the Neural Hawkes Process introduces continuous-time LSTM states [15]. THP, SAHP, and AttNHP apply attention to event histories and irregular timing [28,27,26]; S2P2 develops a continuous-time state-space architecture [3]. These models already represent dependencies across events. We hold a causal attention backbone fixed and test an additional path through observed states. Neither recurrence nor attention alone constitutes the proposed distinction.

Distributions and evaluation targets. Fully neural intensity models parameterize a cumulative hazard [16], while intensity-free models directly specify conditional inter-event distributions [18]. FlexTPP further treats events with mixed discrete and continuous attributes [6]. These formulations preclude a general claim that TPP marks must be categorical. Our quantity head is a numerical regressor, and our time head assigns probability mass to recorded interval bins. Common-head encoder adaptations therefore answer a different comparison from native marked-event likelihoods. Point forecast errors and probabilistic scores also assess different properties [12,8]; we report raw-quantity RMSE and MAE alongside timing NLL at the same quantity-selected checkpoint.

Residual correction. Residual networks establish additive transformation paths [10], and bottleneck adapters apply compact residual modules in transfer learning [11]. ReZero initializes a scalar residual gate at zero [1]. Our zero-initialized local kernels and recurrent output projection preserve the initial base computation through different parameterizations, and the entire event model is trained jointly. The architectural question is whether direct access to recent observed states adds value within a causal attention encoder. The current-only control retains the active parameter count but changes the inputs and operations of the correction path, so the comparison does not isolate parameter count from every possible optimization effect.

Memory and prototype retrieval. Memory networks learn to read from memory representations [20]. Persistent memory adds input-independent key–value entries to attention [21], and product-key memory retrieves learned values from a parameter bank [13].

Titans includes a contextual neural memory whose parameters are updated at test time [2]. In TitanTPP, learned parameters remain fixed during inference; only sequence states and queries depend on the observed prefix. Its persistent attention entries and post-encoder prototypes are separate banks. The retrieval study tests separate normalization of time and quantity queries against a joint learned query with the same summaries and parameter budget. It does not claim the invention of learned memory or top-k retrieval.

3 Method

3.1 Prediction Task and Causal Encoder

Let $\mathcal{H}_j = \{(\Delta_i, q_i)\}_{i \le j}$ be an observed event prefix, where Δ_i is an inter-event interval and $q_i \ge 0$ is an event quantity. The model predicts the recorded interval D_{j+1} and quantity q_{j+1} of the next event. The target position is withheld from the input, and the prediction is read from the last observed position j. Padding is not an event.

Each observed event is embedded as

$$e_i = W_e[\log(1+\Delta_i), \log(1+q_i)]^\top + b_e + p_i, \tag{1}$$

where p_i is a learned position embedding. These two input values are not standardized by training-set moments. The backbone uses Transformer attention [25] in two pre-normalized causal blocks, with hidden width 64, four attention heads, a feed-forward width of 128, and dropout 0.1. Each block also contains 16 learned persistent key–value entries. These entries are distinct from the prototype bank used after the encoder.

Figure 1 shows where the corrections enter. In the first block, the local correction is applied after layer normalization and the query, key, and value projections, before causal attention. The block then completes its attention residual and feed-forward residual. Its output b_i is passed through the recurrent correction before entering the second block. Let h_i denote the resulting second-block output.

3.2 Local and Recurrent History Correction

For a projected query, key, or value sequence X, the local path computes

$$\widetilde{X}_i = X_i + \sum_{\ell=0}^{2} a_{i\ell}\, \kappa_\ell^X \odot X_{s_\ell(i)}. \tag{2}$$

Here $s_0(i) = i$, and $s_1(i), s_2(i)$ are the preceding observed positions in the same correction segment. The mask $a_{i\ell}$ indicates that the corresponding position exists. Each $\kappa_\ell^X \in \mathbb{R}^{64}$ acts channelwise. Applying this three-position operation to queries, keys, and values uses $3 \times 3 \times 64 = 576$ parameters. The operation permits nearby observed states to alter the attention inputs without replacing the attention layer itself.

The recurrent path projects the first-block output into a 54-dimensional state:

$$u_i = Ab_i, \qquad r_i = \mathrm{GRU}(u_i, r_{\mathrm{prev}}), \qquad \widetilde{b}_i = b_i + \mathbf{1}[n_i \ge 2]Br_i. \tag{3}$$

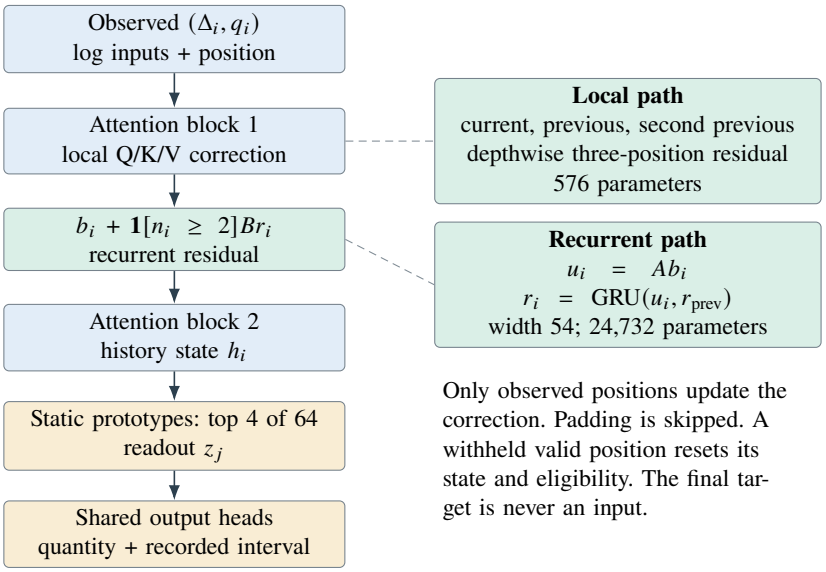


Fig. 1. CNN+GRU54 in TitanTPP. The local correction operates within the first attention block; the recurrent correction operates between blocks. Prototype retrieval and the two output heads follow the encoder. The correction paths contain 25,308 parameters in total.

The GRU follows the gated recurrent formulation [4]; A and B are bias-free projections, and n_i counts observed events in the current correction segment. The residual uses the state *after* processing event i. The recurrent path has 24,732 parameters, including its projections. Its state width is independent of the number of observed events.

Padding neither updates the GRU nor advances the observation count. A valid but withheld position resets the correction state and its eligibility masks. This reset concerns the correction path; it does not change the causal attention mask into a segment-reset mask. The local kernels and recurrent output projection B are initialized to zero. Thus the correction initially leaves the shared encoder computation unchanged, while its input and recurrent parameters can still be learned.

What the controls isolate. The correction-removal variant bypasses both optional paths. The current-only control retains their parameter shapes, initialization, and observation-count masks, but removes direct reads of earlier positions. In Eq. (2), the three bases become X_i, $\tanh(X_i)$, and $\mathrm{GELU}(X_i)$, with the original one-, two-, and three-observation eligibility. For the recurrent component, the current-only control starts from zero at each position and applies two GRU updates using the same current input u_i. Its projected output is used only when $n_i \geq 2$.

The full and current-only paths both contain 25,308 active correction parameters. Their computations nevertheless differ: the current-only control uses nonlinear bases and repeated current-input updates. It also receives b_i, which already summarizes a prefix through attention. Consequently, this comparison isolates a particular explicit

history path against a current-position path; it does not compare a model with history against one without history, or identify the effect of a single lag in isolation.

3.3 Static Prototype Retrieval

After the encoder, a learned bank $M = \{m_k\}_{k=1}^{64}$ supplies a residual to the last observed state h_j. Each slot $m_k \in \mathbb{R}^{64}$ serves as both key and value: its normalized vector is used for scoring, and its raw vector is added to the readout. The slots are fitted with the rest of the model and are not written with observed events at inference time.

Let $\mathrm{norm}(x) = x/\max(\|x\|_2, \epsilon)$ and let S_j contain the four highest-scoring slots. The mean baseline uses cosine scores $s_{jk} = \mathrm{norm}(h_j)^\top \mathrm{norm}(m_k)$ and uniform weights. The weighted variant retains these scores but applies a softmax within the selected four:

$$z_j = h_j + \sum_{k \in S_j} \alpha_{jk} m_k, \qquad \alpha_{jk} = \frac{\exp(s_{jk})}{\sum_{v \in S_j} \exp(s_{jv})}. \tag{4}$$

Mean retrieval instead sets $\alpha_{jk} = 1/4$. Neither variant introduces additional query parameters.

The two learned-query variants receive summaries of the most recent 32 observed positions in the current correction segment, including the current event. Time summaries use standardized $\log(\Delta/s)$; quantity summaries use standardized $\log(1 + q)$, with moments computed from training observations. Each four-dimensional view contains the latest value, mean, population standard deviation, and difference between the latest two eligible values. Nonpositive gaps are excluded from time summaries, whereas zero quantities are retained. The standard deviation and difference require two eligible observations; unavailable summaries are zero. Denote the resulting views by ϕ_t and ϕ_q.

The fixed split-query rule forms

$$v_t = h_j + \phi_t W_t, \qquad v_q = h_j + \phi_q W_q, \tag{5}$$

$$s_{jk}^{\mathrm{fixed}} = \tfrac{1}{2}\,\mathrm{norm}(v_t)^\top \mathrm{norm}(m_k) + \tfrac{1}{2}\,\mathrm{norm}(v_q)^\top \mathrm{norm}(m_k). \tag{6}$$

The two 4×64 projections contain 512 parameters. A budget-matched general-query control uses the same eight summaries, with

$$v_g = h_j + \tfrac{1}{2}[\phi_t, \phi_q] W_g, \qquad s_{jk}^{\mathrm{general}} = \mathrm{norm}(v_g)^\top \mathrm{norm}(m_k), \tag{7}$$

where $W_g \in \mathbb{R}^{8 \times 64}$. Both variants use Eq. (4), temperature one, and zero-initialized query projections. Figure 2 highlights the difference: separate normalization before score averaging versus normalization after the two views have been combined.

The fixed score also has a concatenated dot-product expression:

$$\tfrac{1}{2}[\mathrm{norm}(v_t), \mathrm{norm}(v_q)]^\top [\mathrm{norm}(m_k), \mathrm{norm}(m_k)]. \tag{8}$$

The comparison tests separate query normalization and fixed score mixing. The comparisons share an architecture, not fitted encoder or prototype values: every arm is trained end to end.

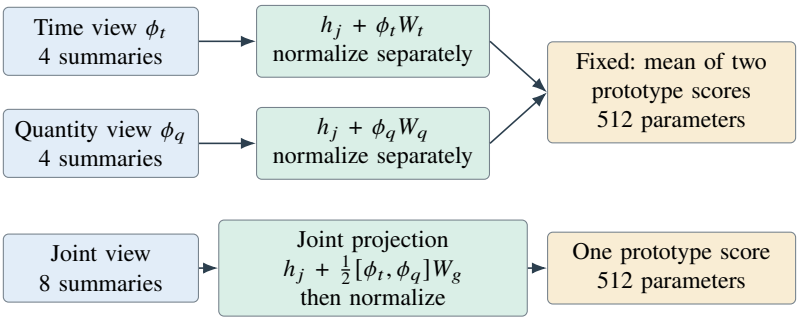


Fig. 2. The matched retrieval comparison. Both rules select four of 64 prototypes and use softmax weights. The fixed rule constrains two normalized views to equal score contributions; the general query learns a joint projection before normalization.

3.4 Shared Output Heads and Training Objective

The quantity head predicts in log space and is evaluated in raw units:

$$\widehat{\ell}_j = \mathrm{softplus}(w_q^\top z_j + b_q), \qquad \widehat{q}_{j+1} = \exp(\widehat{\ell}_j) - 1. \tag{9}$$

No inverse training-set standardization is applied. The time head gives

$$\mu_j = w_\mu^\top z_j + b_\mu, \qquad \sigma_j = 10^{-3} + \mathrm{softplus}(w_\sigma^\top z_j + b_\sigma). \tag{10}$$

For a dataset scale $s > 0$, let $T = s\exp(\mu_j + \sigma_j\varepsilon)$ with $\varepsilon \sim \mathcal{N}(0,1)$ and $F_j(x) = \Phi((\log(x/s) - \mu_j)/\sigma_j)$. The recorded interval is $D = \max(1, \mathrm{round}\, T)$, so its probability mass is

$$p_j(d) = \begin{cases} F_j(1.5), & d = 1, \\ F_j(d + 0.5) - F_j(d - 0.5), & d \geq 2. \end{cases} \tag{11}$$

For Instacart, the recorded interval is capped at 30 days: the interior-bin rule applies for $2 \leq d < 30$, and $p_j(30) = 1 - F_j(29.5)$. Training minimizes the sum of recorded-time negative log-likelihood and squared log-quantity error,

$$\mathcal{L} = -\log p_j(D_{j+1}) + \left(\widehat{\ell}_j - \log(1 + q_{j+1})\right)^2. \tag{12}$$

The quantity weight is one, with no extra large-quantity loss term. All compared arms use these heads and this objective.

4 Experimental Design

4.1 Datasets and Model Selection

Taxi, Intermittent, and RAF Spare Parts form the core study. Instacart provides a supplementary architecture comparison. Table 1 gives the eligible target populations and input windows. Lookback is measured in each dataset's native time bins. The inclusive

Table 1. Target populations and sequence construction. q_{large} is a strict lower bound for the large-quantity subset. Scale s is used by the time distribution.

Dataset	Training	Validation	Cap	Lookback	q_{large}	s
Taxi	38,393	8,268	256	168 hours	3449	1
Intermittent	393,824	86,285	256	520 weeks	187	3
RAF	25,779	6,690	84	84 months	200	6
Instacart	1,991,192	503,733	64	52 days	35	7

window ends at the last observed event; the target is appended before truncation to the sequence cap, which includes that target. Large-quantity thresholds are fixed from the training data. Appendix B gives the event definitions and training distributions; Appendix C examines associations across observed-event lags.

The representative experiments use AdamW [14], learning rate 10^{-3}, weight decay 0.01, batch size 128, and gradient clipping at one. Training lasts at least 40 and at most 300 epochs, with patience 40. Each fit retains the first strictly improving finite minimum of overall raw-quantity RMSE on the full validation set. Ties retain the earlier checkpoint. This rule is shared by the completed MLP study and the new comparisons; neither timing likelihood nor large-quantity error changes the selected epoch.

We evaluate quantity RMSE, quantity MAE, and recorded-time NLL on all validation targets and on the large-quantity subset. All six metrics come from that one checkpoint. Three-seed summaries use the arithmetic mean and sample standard deviation. Time scores are interpreted within a dataset, since the recorded units and distributions differ. Appendix D specifies partitions by quantity and usable history, including how their errors contribute to an aggregate difference.

4.2 Comparisons and Their Scope

Completed MLP correction study. The initial study crosses prototype removal/addition (P_0/P_1) with correction removal/addition (H_0/H_1), giving a 2×2 comparison. It places an MLP correction between the first and second attention blocks, with no local Q/K/V correction; prototype retrieval follows the second block. Thus it differs from both correction paths in CNN+GRU54. Eight bias-free branches each map the concatenated current and immediately preceding observed states through $128\to8\to64$, and their outputs are averaged. The branches become eligible after more than 1, 2, 4, 8, 16, 32, 64, or 128 observations. These are eligibility thresholds, not eight different history lags.

The correction contains 12,288 parameters; the 64-slot prototype bank contains 4,096. Two additional controls test alternatives to these components. P_1H_C replaces the correction with eight current-state branches of width 12, matching its parameter count. P_FH_1 replaces the prototype residual with a bias-free $64\to32\to64$ feed-forward residual, also using 4,096 parameters. All six arms retain the backbone's persistent attention entries. The six arms, three datasets, and seeds 42, 52, and 62 produce 54 completed fits.

These budget controls have limits. The feed-forward substitute differs in activation and initialization from prototype retrieval. Equal parameter counts do not imply equal

computation. On RAF, the sequence cap prevents the eighth MLP branch from becoming eligible, leaving seven potentially active branches. These details constrain what can be attributed to the nominal budgets.

Representative correction and retrieval studies. CNN+GRU54 is compared with correction removal and the current-only control on each core dataset and seed: 27 fits. The retrieval comparison adds weighted, fixed, and general-query variants, with the nine full-correction models reused as the mean baseline. This requires 27 further fits, rather than a second training of the same mean models. The matched general query has 512 added parameters; it is distinct from the 521-parameter control used in the earlier exploratory retrieval pilot.

For a dataset and seed, all comparison arms run on the same GPU and software environment. Taxi and RAF are assigned to RTX 5080; Intermittent to RTX 5090. A preliminary implementation check compared CNN+GRU54 with mean retrieval on RTX 5080: 83 bitwise checks covered initialization, synthetic sequences, and the first three updates on a shared Taxi training batch. This establishes a limited implementation baseline, not equivalence of past training trajectories or checkpoints obtained on another GPU.

On Instacart, the supplementary study compares MLP8, MLP12, and CNN+GRU54 across the same three seeds, adding nine fits. It asks about architecture behavior on a fourth dataset; it does not replace the correction or retrieval controls. Together, these studies specify 63 new fits. Their completed comparisons will determine the representative model's evidence, independently of the MLP findings below.

Decision criteria and computational cost. The correction criterion requires the full model to have lower overall quantity RMSE than both controls in every seed on the same two or more core datasets. The fixed-retrieval criterion requires corresponding improvements over mean, weighted, and the general-query control. These are descriptive consistency criteria, not significance tests.

Retrieval must also meet a predeclared cost bound relative to mean retrieval on the same GPU. With batch 128, sequence lengths 64 and 256, five warm-up iterations and twenty measured iterations, the allowed increases are 20% for inference time, 30% for training-step time, and 20% for peak allocation. Both lengths must pass. Results from arms above these bounds remain part of the comparison.

5 Results

5.1 Quantity Gains in the Completed MLP Study

Table 2 reports the complete six-arm quantity comparison. Adding the MLP correction reduces overall RMSE in all 18 paired comparisons: three datasets, three seeds, and two prototype settings. Without prototypes, the reductions in the means are 13.04% on Taxi, 9.01% on Intermittent, and 0.77% on RAF. With prototypes, the corresponding reductions are 14.11%, 3.28%, and 0.85%. The direction is consistent, but the magnitude is strongly dataset-dependent.

Table 2. Completed MLP study: overall validation quantity RMSE (mean ± sample SD over three seeds; lower is better). Bold marks the lowest mean within each dataset. These are not CNN+GRU54 results.

Arm	Taxi	Intermittent	RAF
P_0H_0	88.977 ± 0.501	1.777 ± 0.065	34.242 ± 0.147
P_1H_0	90.509 ± 0.521	1.721 ± 0.045	34.230 ± 0.127
P_0H_1	**77.377** ± 1.822	**1.617** ± 0.009	33.979 ± 0.045
P_1H_1	77.735 ± 0.681	1.664 ± 0.009	**33.938** ± 0.071
P_FH_1	79.255 ± 1.230	1.759 ± 0.124	34.125 ± 0.074
P_1H_C	88.846 ± 1.267	1.728 ± 0.061	34.273 ± 0.327

The parameter controls provide a more selective result. P_1H_1 beats both P_FH_1 and P_1H_C in all three seeds on Taxi and RAF. On Intermittent, it beats each control in only two of three seeds. Thus the original MLP study satisfies its two-dataset consistency criterion, while leaving an exception on Intermittent. The large Taxi gap against P_1H_C is compatible with useful explicit previous-state input in this MLP architecture; the control does not establish which part of the CNN+GRU path would account for such an effect.

Prototype addition yields a different pattern. With the correction already present, P_1H_1 has 0.46% higher mean RMSE than P_0H_1 on Taxi and 2.92% higher RMSE on Intermittent. On RAF it improves the mean by 0.12%. The completed evidence therefore favors explicit correction more consistently than prototype addition. Choosing a different best variant for each dataset would conceal this distinction and would not define a single representative model.

5.2 Timing, Large Quantities, and Cost

Quantity gains are not uniform gains across outputs. On Taxi, adding correction raises overall time NLL from 0.818 to 1.253 without prototypes and from 0.725 to 1.229 with prototypes. These are increases of 53.17% and 69.64% in the means; timing NLL worsens in every seed under both settings. The checkpoints were selected for quantity RMSE, so this is a tradeoff at the quantity-selected model rather than a comparison of separately optimized timing checkpoints.

Large-quantity behavior also qualifies the overall result. The Taxi full model has large-quantity RMSE 376.660, compared with 366.709 for its feed-forward prototype substitute, despite lower overall RMSE. On RAF, adding correction slightly reduces overall RMSE but increases large-quantity MAE by 1.27% without prototypes and 0.94% with prototypes. Its large-quantity time NLL rises by 3.64% and 3.85%, respectively. The large subsets contain only 79 Taxi targets and 50 RAF targets, compared with 1,141 on Intermittent. Appendix A retains all six metrics, including these unfavorable outcomes.

The historical technical checks do not show that prototype retrieval is cheaper than its parameter-matched substitute. On RTX 5080, the measured step time for P_1H_1 is 36.46 ms versus 25.34 ms for P_FH_1 at length 256, and 14.15 ms versus 11.49 ms at length 84. On RTX 5090 at length 256, the corresponding times are 23.11 and 17.74 ms.

These checks use the median of three synthetic optimizer steps, without discarded warm-up, and include gradient-copy and verification work. They are limited technical measurements, not pure throughput or inference benchmarks, and do not implement the new retrieval cost protocol.

5.3 Evidence Available for the Representative Architecture

The CNN+GRU correction and matched retrieval comparisons do not yet have completed results in the evidence used here. Their implementation and experimental controls are specified in Sections 3 and 4, but the MLP results cannot establish their effects.

The archived observation of 8 October 2026, 14:38 KST contains one completed Instacart fit: MLP8, seed 52, on RTX 5090. It selected epoch 15 and stopped after epoch 55, with overall quantity RMSE 5.884, MAE 3.994, and time NLL 2.812. For the 6,036 large-quantity targets, the same checkpoint gives RMSE 24.665, MAE 22.088, and time NLL 2.900. This is a single-fit observation, not a three-seed architecture comparison; final collection of the complete stage remains outstanding. It cannot rank MLP8, MLP12, and CNN+GRU54.

6 Discussion

The completed study gives a specific empirical motivation for explicit correction. Quantity RMSE improves consistently when the MLP path is added, and the full model beats the two budget controls in every seed on two datasets. At the same time, prototype addition is not consistently helpful, and the largest correction gains coincide with a timing penalty. These results argue for separating the two mechanisms and for keeping timing and large-quantity errors visible alongside the primary metric.

The representative architecture changes the correction mechanism substantially. Its local operation reads three observed positions, while its GRU carries a recurrent state across the observed prefix. A successful full-versus-current-only comparison would support that combined path, not a claim that one preceding state alone explains the gain. Equal active parameter counts narrow the capacity explanation but do not remove differences in operations, depth, or optimization. Component-level attribution would require further controls.

The retrieval hypothesis is narrower still. Fixed split-query scoring can be expressed as a concatenated dot product. A benefit over the matched general query would support the tested separate-normalization constraint, not the invention of a fundamentally new memory mechanism. A benefit over mean alone would be insufficient, because both similarity weighting and general query learning are plausible alternatives. Earlier exploratory observations informed this hypothesis, making the new comparison a development study rather than independent confirmation.

All performance reported here comes from validation evidence used during research. An independently unexposed evaluation is not yet available. The present comparisons also do not establish superiority over external models such as S2P2: encoder adaptations, output heads, training conditions, and selection rules must be aligned before such a comparison can support a claim. Three seeds and directional consistency describe the

current evidence; they do not establish statistical significance or performance across all event-forecasting settings.

7 Conclusion

We examined explicit history correction within a causal attention model. The completed MLP analysis finds consistent quantity improvements across paired runs, together with timing and large-quantity costs that limit a broader performance claim. Its prototype results are less consistent. The CNN+GRU architecture and matched controls carry these observations into two focused questions: whether an explicit local and recurrent path adds value, and whether separate time and quantity queries improve prototype retrieval. Those comparisons, followed by an unexposed evaluation, will determine how far the present evidence extends.

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A Full Validation Metrics for the MLP Study

All entries in Table 3 are means and sample standard deviations over seeds 42, 52, and 62. Each fit contributes metrics from its first minimum-overall-quantity-RMSE checkpoint. The large subsets use the strict training-derived thresholds in Table 1. Values are rounded to three decimals except for time NLLs below 0.001, which use scientific notation with the standard deviation on the line beneath the mean. No metric in this table comes from the representative CNN+GRU experiment.

Table 3. Completed MLP study: all six validation metrics (mean ± sample SD). Lower is better in every column. The large-quantity subset is shown beside the full population rather than used for independent checkpoint selection.

	All targets			Large-quantity targets		
Arm	RMSE	MAE	Time NLL	RMSE	MAE	Time NLL
Taxi: 8,268 targets; 79 large						
P_0H_0	88.977 ± 0.501	28.313 ± 0.096	0.818 ± 0.191	419.317 ± 53.142	336.399 ± 45.621	1.21×10^{-30} $\pm2.09\times10^{-30}$
P_1H_0	90.509 ± 0.521	28.755 ± 0.453	0.725 ± 0.067	431.690 ± 19.168	346.009 ± 21.136	6.36×10^{-17} $\pm1.10\times10^{-16}$
P_0H_1	77.377 ± 1.822	25.180 ± 0.491	1.253 ± 0.125	385.604 ± 27.258	311.795 ± 29.080	9.77×10^{-78} $\pm1.69\times10^{-77}$
P_1H_1	77.735 ± 0.681	25.040 ± 0.146	1.229 ± 0.160	376.660 ± 10.552	296.101 ± 11.577	5.51×10^{-62} $\pm9.54\times10^{-62}$
P_FH_1	79.255 ± 1.230	25.475 ± 0.205	1.155 ± 0.373	366.709 ± 26.906	287.777 ± 23.170	1.58×10^{-43} $\pm2.74\times10^{-43}$
P_1H_C	88.846 ± 1.267	28.422 ± 0.295	1.001 ± 0.322	398.429 ± 29.267	314.315 ± 23.846	1.82×10^{-25} $\pm3.15\times10^{-25}$
Intermittent: 86,285 targets; 1,141 large						
P_0H_0	1.777 ± 0.065	0.751 ± 0.042	0.366 ± 0.016	8.121 ± 0.462	6.406 ± 0.382	0.035 ± 0.046
P_1H_0	1.721 ± 0.045	0.735 ± 0.048	0.371 ± 0.050	8.000 ± 0.165	6.163 ± 0.145	1.32×10^{-4} $\pm2.28\times10^{-4}$
P_0H_1	1.617 ± 0.009	0.669 ± 0.015	0.384 ± 0.041	7.510 ± 0.441	5.561 ± 0.283	0.004 ± 0.007
P_1H_1	1.664 ± 0.009	0.722 ± 0.009	0.370 ± 0.059	7.711 ± 0.125	5.986 ± 0.290	0.016 ± 0.028
P_FH_1	1.759 ± 0.124	0.745 ± 0.105	0.356 ± 0.068	8.911 ± 0.805	6.481 ± 0.575	0.387 ± 0.645
P_1H_C	1.728 ± 0.061	0.710 ± 0.023	0.551 ± 0.350	8.315 ± 0.473	6.541 ± 0.402	9.908 ± 17.131
RAF: 6,690 targets; 50 large						
P_0H_0	34.242 ± 0.147	9.270 ± 0.057	3.592 ± 0.112	326.556 ± 6.460	279.427 ± 7.680	3.455 ± 0.150
P_1H_0	34.230 ± 0.127	9.261 ± 0.062	3.596 ± 0.112	326.405 ± 6.068	279.348 ± 7.255	3.447 ± 0.145
P_0H_1	33.979 ± 0.045	9.259 ± 0.099	3.554 ± 0.078	327.105 ± 1.589	282.972 ± 2.416	3.581 ± 0.220
P_1H_1	33.938 ± 0.071	9.266 ± 0.092	3.552 ± 0.081	326.227 ± 2.112	281.968 ± 2.766	3.580 ± 0.223
P_FH_1	34.125 ± 0.074	9.189 ± 0.031	3.470 ± 0.029	329.715 ± 0.401	285.503 ± 0.716	3.336 ± 0.068
P_1H_C	34.273 ± 0.327	9.291 ± 0.084	3.548 ± 0.075	327.082 ± 4.843	282.363 ± 6.635	3.445 ± 0.144

B Data definitions and training distributions

Taxi events record pickup counts in active spatial-cell hours. Intermittent events contain positive site–part order quantities, and RAF events contain positive monthly spare-parts demands. Instacart aggregates product rows for each user on the same recorded activity day, so its quantity measures basket size rather than physical units. Empty calendar periods extend the interval between events instead of creating zero-quantity targets. The recorded units are hours, weeks, months, and days, respectively. Instacart intervals are top-coded at 30 days.

Table 4 describes the frozen training populations. Quantity summaries include every training event, whereas gap and usable-history summaries exclude each series' first event because it has no predecessor for prediction. Quantiles select the nearest observed value. Each summary weights events or prediction targets equally within its population, rather than weighting series equally.

Usable history counts the observed predecessor events retained for a target after both window restrictions. The inclusive recorded-time window ends at the last observed event. Appending the target before sequence truncation reserves one position for it, so at most the sequence cap minus one observed events remain. Neither the target nor padding contributes to usable history. Table 1 specifies the windows and caps. The train descriptions therefore characterize the same input contract across the MLP and CNN+GRU studies, although those architectures process the retained events differently.

Table 4. Training-only event characteristics. Quantity summaries include all training events; gap, usable-history, and adjacent-pair summaries describe targets after excluding the first event of each series. Correlation is defined in Appendix C.

Characteristic	Taxi	Intermittent	RAF	Instacart
Time unit	Hour	Week	Month	Day
Training series	131	5,000	5,000	206,209
Training event rows	38,524	398,824	30,779	2,197,401
Training targets	38,393	393,824	25,779	1,991,192
Quantity: median	7	2	2	8
Quantity: 95th percentile	1,562	46	60	25
Quantity: 99th percentile	3,449	187	200	35
Quantity: maximum	6,322	477	2,062	175
Recorded gap: median	1	3	6	7
Recorded gap: 95th percentile	5	5	23	30
Usable history: median	129	40	3	5
Usable history: 95th percentile	168	144	8	14
History ≤ 7 events (%)	2.41	8.89	93.58	72.13
Within-series log correlation	0.749	0.928	-0.105	0.037
Equal adjacent quantities (%)	20.33	61.99	33.30	10.24

Figure 3 indicates substantial differences in available history. The median spans 129 observed events on Taxi and 40 on Intermittent, compared with three on RAF and

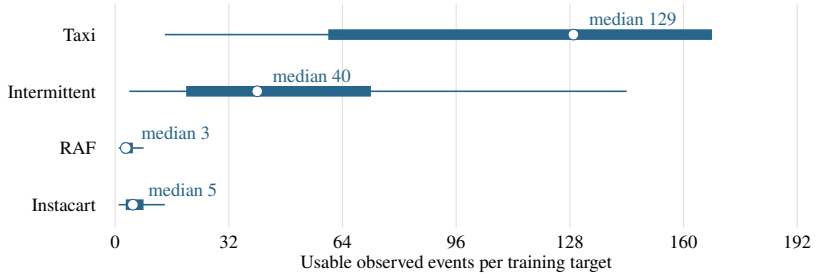


Fig. 3. Available training history under the frozen input windows. Thin lines span the 5th–95th percentiles, thick lines the 25th–75th percentiles, and circles mark medians. These target-weighted distributions characterize input availability, not model performance.

five on Instacart. Thus an equal nominal sequence cap would not imply equal evidence per prediction. These training distributions describe the input populations; they neither measure which events a trained model retains nor determine its forecasting error.

C Event-lag association in the training data

For series e and event lag ℓ, pair $x_{e,j}^{(\ell)} = \log(1 + q_{e,j-\ell})$ with $y_{e,j}^{(\ell)} = \log(1 + q_{e,j})$ whenever both events belong to that training series. Each lag counts observed events in sequence order. It does not denote elapsed calendar time. The descriptive calculation includes the complete training sequence, without the model's recorded-time window or sequence-length cap.

Let $\bar{x}_e^{(\ell)}$ and $\bar{y}_e^{(\ell)}$ be the separate means of the predecessor and successor members of those pairs. The statistic is

$$r_\ell = \mathrm{Corr}\left(x_{e,j}^{(\ell)} - \bar{x}_e^{(\ell)},\ y_{e,j}^{(\ell)} - \bar{y}_e^{(\ell)}\right), \tag{13}$$

where correlation pools the centered pairs across series. Centering removes between-series differences in mean log quantity, but each pair still contributes to the pooled sums. This statistic is consequently different from an average of series-specific correlations. The lag-one column corresponds to the adjacent log correlation in Table 4.

Taxi's association declines with event lag, while Intermittent retains positive association over the displayed range. RAF supplies comparatively few long-lag pairs, and Instacart's correlations remain close to zero. Counts in Figure 4 matter because the available series and pairs change with lag. A series with one pair contributes zero to both centered sums of squares. Constant pairs also provide no variation, and a zero pooled denominator leaves the correlation undefined. RAF's negative values therefore describe these centered samples without establishing a general law of negative dependence.

An archived sensitivity calculation restricts every lag to the same series with at least 20 training events. This cohort retains 131 Taxi series, 4,968 Intermittent series, and 31,020 Instacart series; RAF supplies none. Instacart's lag-one and lag-two correlations

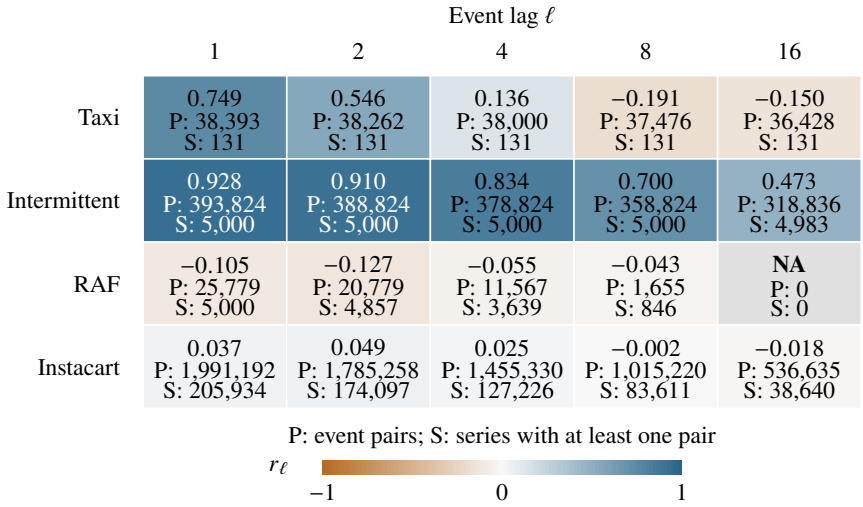


Fig. 4. Train-only within-series log-quantity correlation at event lags 1, 2, 4, 8, and 16. Each cell reports r_ℓ, the number of event pairs (P), and the number of contributing series (S). All cells share the color scale $[-1, 1]$; gray NA denotes no available pairs. Pair counts include constant and single-pair series, which contribute zero to one or both centered sums of squares. Pairs cover complete training sequences without input-window or sequence-cap restrictions.

increase to 0.074 and 0.092 in this cohort. The change cautions against treating the full-population profile as invariant to series composition.

Event-lag association provides a descriptive motivation for examining local and recurrent history paths. It does not measure the contribution of either path. CNN+GRU54 processes contextual states within the retained input window, whereas Eq. (13) describes unconditional log-quantity pairs across complete training sequences. Neither the sign nor the magnitude of r_ℓ establishes a correction gain, a preferred recurrent width, or an architectural ranking. Those claims depend on the matched validation comparisons specified in the main text.

D Diagnostic partitions and aggregation

D.1 Quantity and usable-history partitions

Stratification distinguishes a change in errors within a group from a change in that group's contribution to the aggregate. The quantity boundaries are the training-derived 50th, 90th, 95th, and 99th percentiles, namely $(7, 686, 1562, 3449)$ for Taxi, $(2, 31, 46, 187)$ for Intermittent, $(2, 30, 60, 200)$ for RAF, and $(8, 20, 25, 35)$ for Instacart. Each upper boundary belongs to the lower bin, so the last bin agrees with the strict large-quantity threshold in Table 1.

The stored history partitions divide Taxi, Intermittent, and RAF at 64 and 128 usable events, and Instacart at 1, 3, 7, 15, and 31. They permit within-dataset diagnosis, although

they do not define identical groups across datasets. Their resolution also depends on the sequence cap. Empty strata remain explicit and receive no estimated error. For a fixed dataset, every comparison arm and seed must share target membership and partition boundaries.

All strata inherit the checkpoint selected by minimum overall validation quantity RMSE. Neither a favorable bin nor a timing score changes that selection. A prespecified control remains fixed across bins; if an external comparator is selected by aggregate validation RMSE, that same comparator also remains fixed. These rules prevent a separate best-model choice within each subgroup. The definitions below specify the diagnostic procedure and introduce no additional CNN+GRU performance results.

D.2 Per-seed errors and additive contributions

For model m, seed s, and a nonempty stratum b containing N_b targets, define

$$\mathrm{SSE}_{m,s,b} = \sum_{i \in b} (\widehat{q}_{m,s,i} - q_i)^2, \tag{14}$$

$$\mathrm{MAE}_{m,s,b} = \frac{1}{N_b} \sum_{i \in b} |\widehat{q}_{m,s,i} - q_i|, \qquad \mathrm{RMSE}_{m,s,b} = \sqrt{\frac{\mathrm{SSE}_{m,s,b}}{N_b}}. \tag{15}$$

Compute each metric within a seed before averaging over $\mathcal{S} = \{42, 52, 62\}$. Sample standard deviations describe the dispersion of these seed-level metrics. The count N_b refers to targets in one seed, rather than repeated copies of the same target across seeds.

Let M denote the focal model, C its fixed comparator, and $N = \sum_b N_b$. The contributions of bin b to the aggregate differences are

$$C_b^{\mathrm{MSE}} = \frac{1}{|\mathcal{S}|N} \sum_{s \in \mathcal{S}} (\mathrm{SSE}_{M,s,b} - \mathrm{SSE}_{C,s,b}), \tag{16}$$

$$C_b^{\mathrm{MAE}} = \frac{N_b}{N|\mathcal{S}|} \sum_{s \in \mathcal{S}} (\mathrm{MAE}_{M,s,b} - \mathrm{MAE}_{C,s,b}). \tag{17}$$

Summing these contributions across bins recovers the mean of the seeds' overall MSE or MAE differences. Negative values favor M. The MSE sum is not the difference between squared mean RMSEs, and binwise RMSE differences do not supply an additive decomposition. Empty bins contribute zero to the aggregate through their zero target count, without an estimated bin metric.

Because each contribution includes target share, a small group with large errors can offset gains across a larger group. Reporting both within-bin errors and contributions distinguishes these cases without attributing the difference to a single data characteristic. A model-specific decomposition requires that model's selected-checkpoint validation records. Historical MLP decompositions cannot substitute for CNN+GRU54 results.